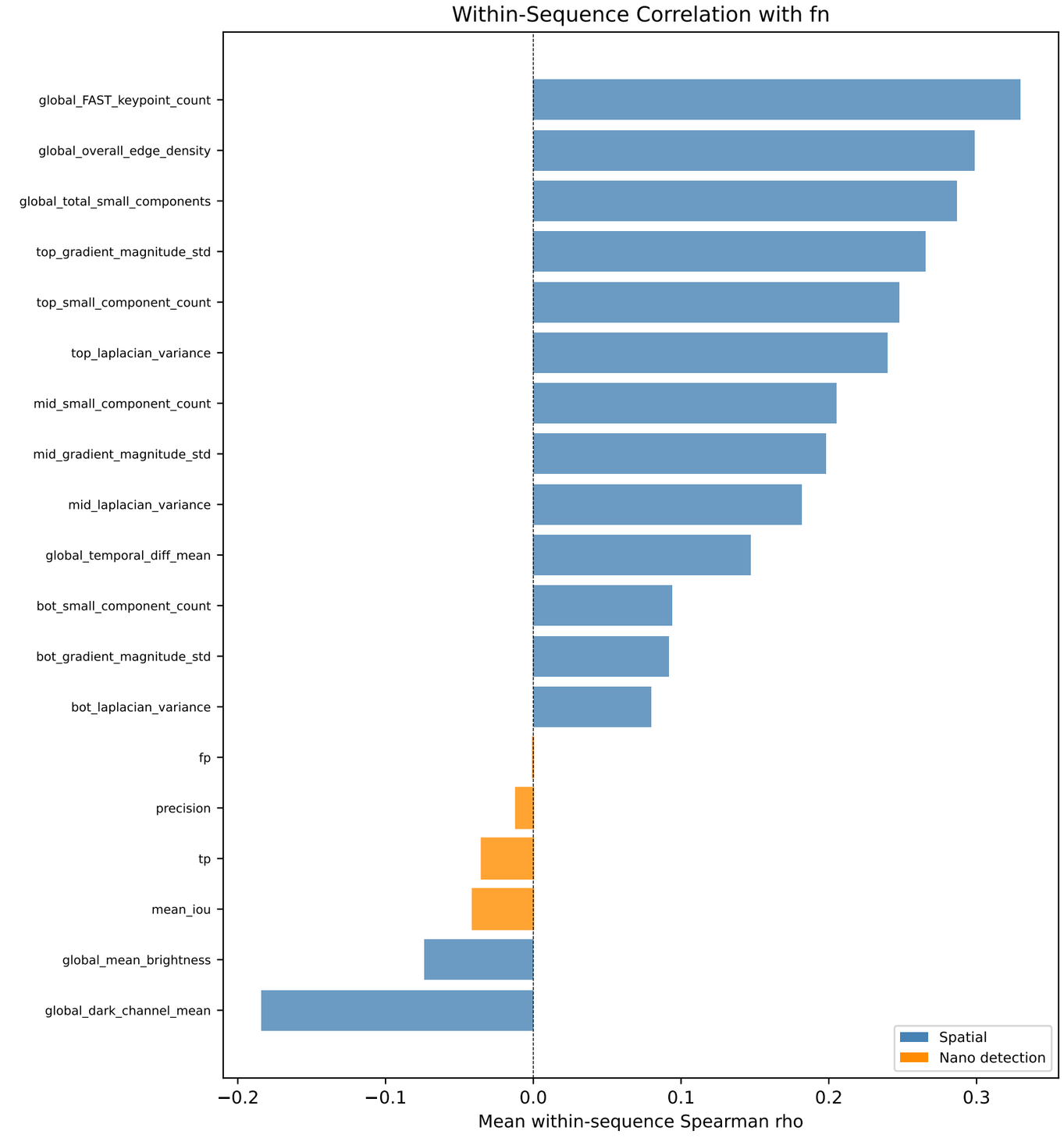
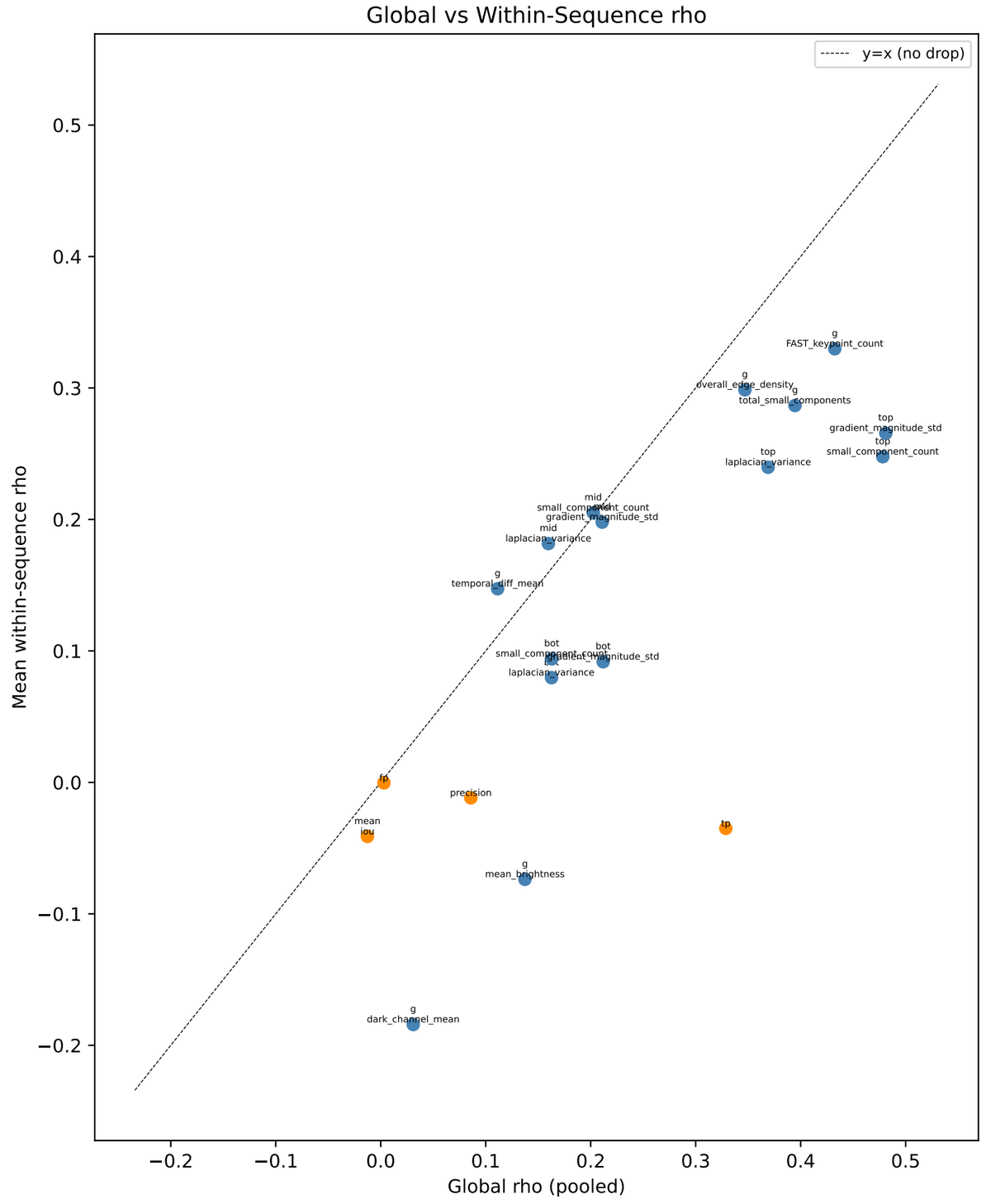
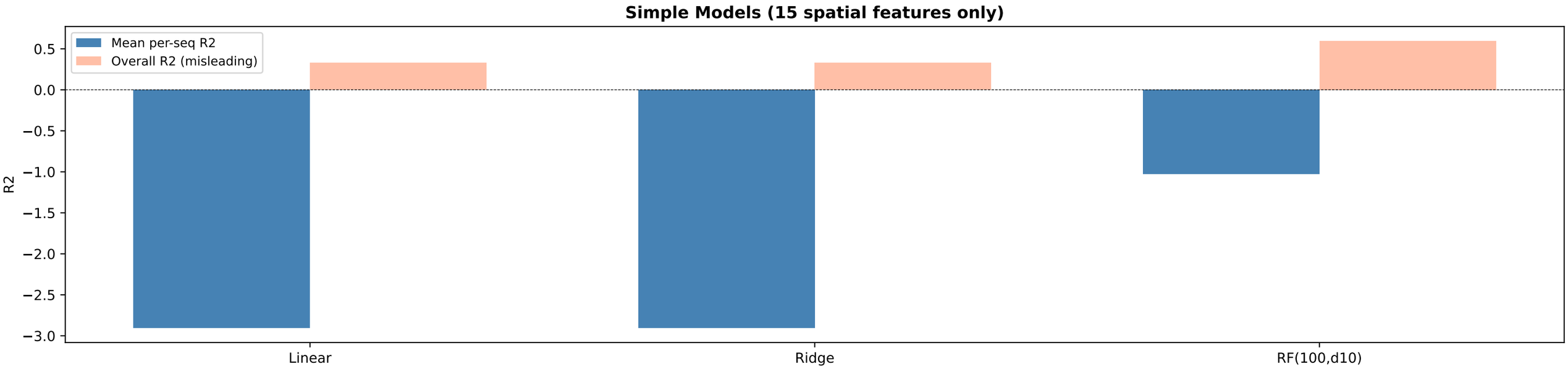
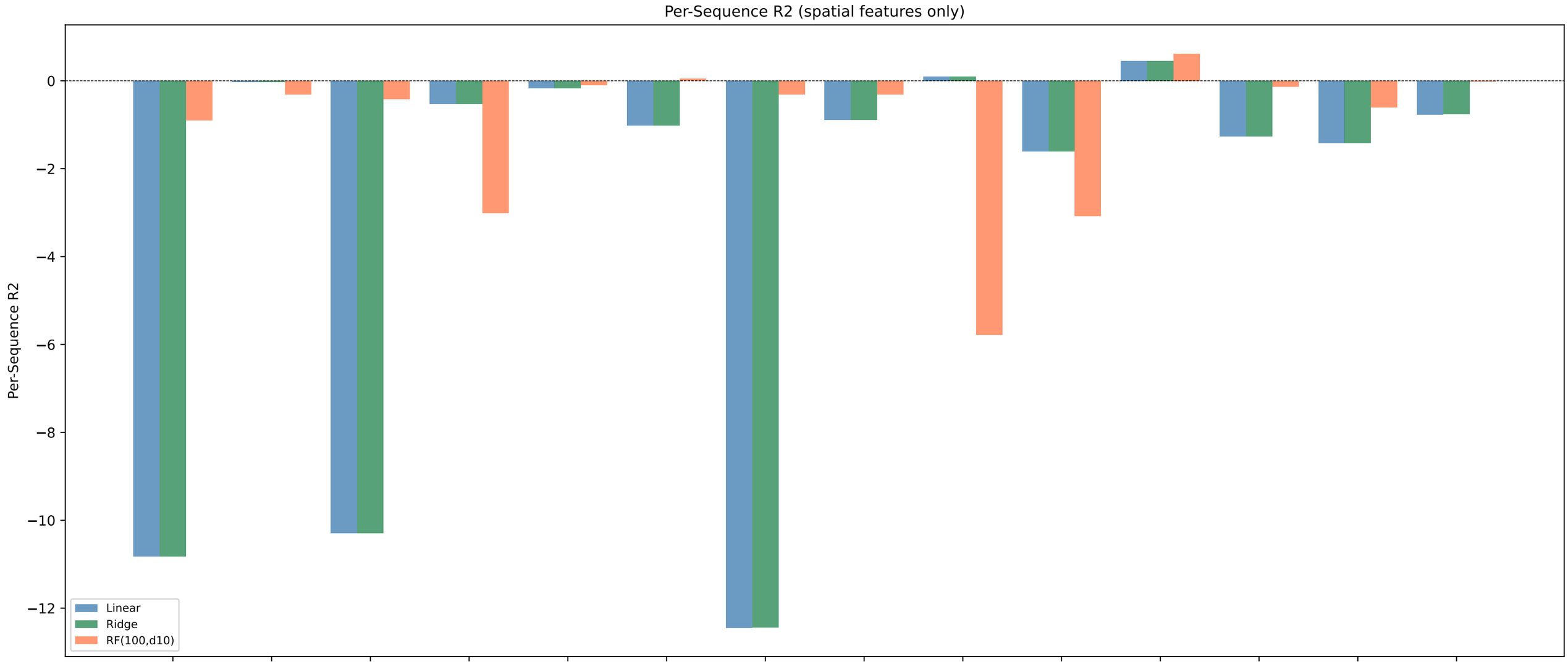
Page 1: Within-Sequence Feature Correlation





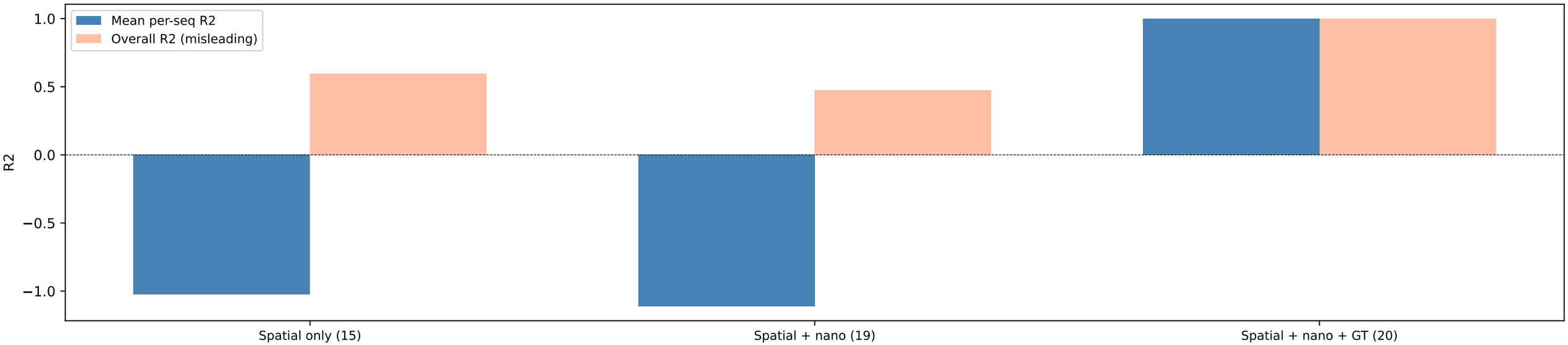
Page 2: Simple Per-Frame Models (No LSTM, No Temporal)



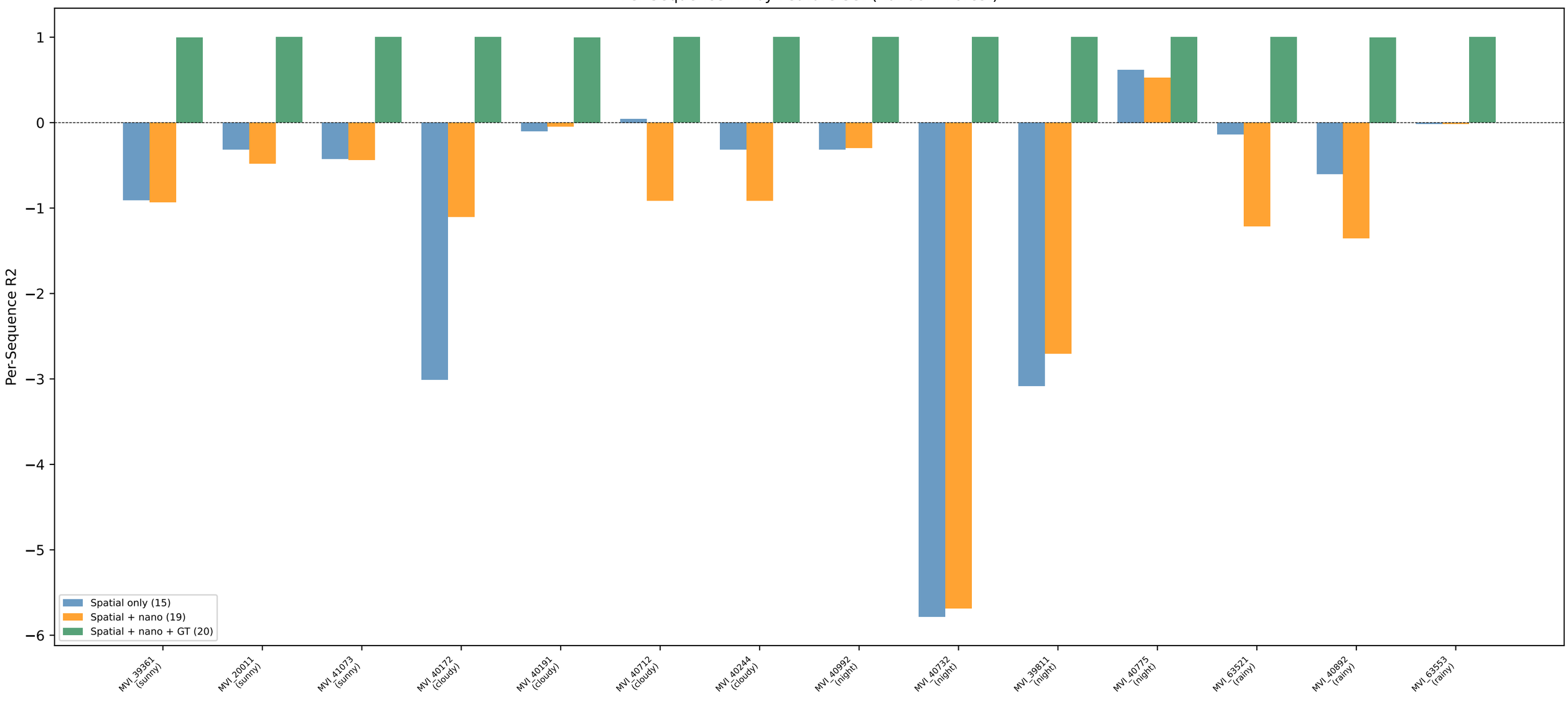


Page 3: Detection Features Impact

Does Adding Nano@320 Detection Output Help?



Per-Sequence R2 by Feature Set (Random Forest)



Page 4: Diagnostic Summary

```
DIAGNOSTIC SUMMARY
============================================================

TEST 1: Within-Sequence Correlation
  Mean |global rho| (spatial features): 0.260
  Mean within-seq rho (spatial features): 0.160
  Best global:  top_gradient_magnitude_std = 0.481
  Best within:  global_FAST_keypoint_count = 0.330
  Signal drop: 0.099

TEST 2: Simple Per-Frame Models (spatial only)
  Best model: RF(100,d10)
  Mean per-seq R2: -1.0245
  (LSTM intra-seq R2 was: 0.200)

TEST 3: Detection Features Impact (Random Forest)
  Spatial only (15):           mean per-seq R2 = -1.0245
  Spatial + nano (19):         mean per-seq R2 = -1.1119
  Spatial + nano + GT (20):    mean per-seq R2 = 0.9996
  Nano detection lift: -0.0875
  Oracle (+ num_gt) lift: +2.0241

============================================================
DIAGNOSIS:
  Within-seq rho 0.1-0.2 => weak but present intra-seq signal.
  Features are the primary bottleneck, model is secondary.
  Adding nano detections barely helps (+-0.087).
  Even simple models fail => per-frame fn is fundamentally
  hard to predict from image features alone.
```